

# Power Series and Taylor Series



## Radius and interval of convergence

- 1 Find the radius of convergence of  $\sum_{n=1}^{\infty} \frac{x^n}{n}$ .
  - 2 Find the radius of convergence of  $\sum_{n=0}^{\infty} \frac{x^n}{n!}$ .
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## Taylor and Maclaurin series

- 3 Find the first three nonzero terms of the Maclaurin series for  $f(x) = e^x$ , using  $f^{(n)}(0) = 1$  for all  $n$ .
- 4 Find the third-degree Taylor polynomial for  $f(x) = \cos x$  centered at  $x = 0$ .
- 5 Use the Maclaurin series  $\sin x = x - \frac{x^3}{6} + \frac{x^5}{120} - \dots$  to approximate  $\sin(0.2)$  using the first two nonzero terms, to four decimal places.